

Instructions: Print this out and set aside 50 minutes for you to take this mock exam **with no notes nor a calculator present**. Each question should take approximately 4.5 minutes to answer. **Good luck and have fun!**

1. Simplify each expression completely. No negative exponents should remain.

(a) $(-4x^2)^3(-6x^{-3})^3$

(b) $\left(\frac{-2x^{-2}y^4z^3}{2x^2y^2z^4}\right)^{-2} \cdot (xy)^0$

2. Expand and simplify the expression by combining like terms. Do not factor your final expression.

$$(5x + 1)^2 - (5x + 1)$$

3. Find the quotient and remainder of the following polynomial long division problem:

$$(2x^3 + x^2 + 3x + 2) \div x - 3.$$

4. Factor the expressions **completely**

(a) $16x^2y - 4y$

(b) $4x^3 - 16x^2 - x + 4$

5. Solve the following quadratic equation using factoring:

$$x^2 + 10x - 9 = 16.$$

6. A ball launched upward from the top off a 64-foot building that is modeled by the following quadratic equation:

$$h(t) = -16t^2 + 48t + 64.$$

At what time t (in seconds) does the ball hit the ground?

7. Find the domain of the function and express your answer in **interval notation**:

$$\sigma(x) = \frac{2}{x^2 + 8x + 15}.$$

8. Divide/multiply and simplify the following expression completely:

$$\frac{x^2 - 25}{x^2 - 8x + 15} \div \frac{x - 5}{x - 3}$$

9. Add and simplify the following rational expressions completely into a single a fraction:

$$\frac{x^2}{x + 3} + \frac{x}{x^2(x - 3)}.$$

10. Solve for x and state any extraneous solutions

$$1 + \frac{1}{y - 5} = \frac{3}{(y - 5)(y - 2)}.$$

11. Working together, it takes two computers 30 minutes to process a data set. If it takes the faster computer 50 minutes to do the job on its own, how long will it take the slower computer to do the job on its own?